

Linear Programing Problem : Structure of linear program model, Assumptions, Advantages, Limitations, General mathematical model, Guidelines for formulation of linear programming model, Formulation of problems, graphical method.

I. MEANING / DEFINITION OF LPP

Linear Programming is a mathematical technique useful for allocation of ‘scarce’ or ‘limited’ resources, to several competing activities on the basis of a given criterion of optimality – **J K Sharma**.

Linear Programming is a planning technique that permits some objective functions to be minimized or maximized within the frame work of given situational restrictions – **William M Fox**.

II. STRUCTURE OF LINEAR PROGRAM MODEL

Four components of any LP model are :

1. Decision variables,
2. Objective function,
3. Constraints, and
4. Non-negativity

1. **Decision variables (activities)** : The evaluation of various courses of action (alternatives) and select the best to arrive at the optimal value of objective function, is guided by the nature of objective function and availability of resources. For this, certain activities (also called *decision variables*) usually denoted by $x_1, x_2, \dots, x_n$ are conducted. The value of these variables (activities) represents the extent to which each of these is performed.

For example : In a product-mix manufacturing problem, an LP model may be used to determine units of each of the products to be manufactured by using limited resources such as personnel, machinery, money, material, etc.

The value of certain variables may or may not be under the decision-maker’s control.

- If values are under the control of the decision-maker, then such variables are said to be **controllable**,
- otherwise they are said to be **uncontrollable**. These decision variables, usually interrelated in terms of consumption of resources, require simultaneous solutions. In an LP model all decision variables are continuous, controllable and nonnegative. That is, $x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0$.

2. **The objective function** : The objective function of each LP problem is expressed in terms of decision variables to optimize the criterion of optimality (also called *measure-of-performance*) such as profit, cost, revenue, distance etc. In its general form, it is represented as:
Optimize (Maximize or Minimize) $Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$,
where Z is the measure-of-performance variable, which is a function of $x_1, x_2, \dots, x_n$. Quantities $c_1, c_2, \dots, c_n$ are parameters that represent the contribution of a unit of the respective variable

$x_1, x_2, \dots, x_n$ to the measure-of-performance Z . The optimal value of the given objective function is obtained by the graphical method or simplex method.

3. **The constraints :** There are always certain limitations (or constraints) on the use of resources, such as: labour, machine, raw material, space, money, etc., that limit the degree to which an objective can be achieved. Such constraints must be expressed as linear equalities or inequalities in terms of decision variables. The solution of an LP model must satisfy these constraints.
4. **Non – Negativity :** The values or decision variables should possess positive or zero.

III. ASSUMPTIONS OF AN LP MODEL

1. **Certainty :** In LP models, it is assumed that all its parameters such as: availability of resources, profit (or cost) contribution per unit of decision variable and consumption of resources per unit of decision variable must be known and constant.
2. **Additivity :** The value of the objective function and the total amount of each resource used (or supplied), must be equal to the sum of the respective individual contribution (profit or cost) of the decision variables.
For example : The total profit earned from the sale of two products A and B must be equal to the sum of the profits earned separately from A and B. Similarly, the amount of a resource consumed for producing A and B must be equal to the total sum of resources used for A and B individually.
3. **Linearity (or proportionality):** The amount of each resource used (or supplied) and its contribution to the profit (or cost) in objective function must be proportional to the value of each decision variable.
For example : If production of one unit of a product uses 5 hours of a particular resource, then making 3 units of that product uses $3 \times 5 = 15$ hours of that resource.
4. **Divisibility (or continuity):** The solution values of decision variables are allowed to assume continuous values. For instance, it is possible to collect 6.254 thousand litres of milk by a milk dairy and such variables are divisible. But, it is not desirable to produce 2.5 machines and such variables are not divisible and therefore must be assigned integer values. Hence, if any of the variable can assume only integer values or are limited to discrete number of values, LP model is no longer applicable.
5. **Finiteness :** An optimum solution cannot be computed in the situations where there are an infinite number of alternative activities and resource restrictions.
6. **Optimality :** In LPP, the maximum profit solution or the minimum cost solution always occurs at a corner point of the set of feasible solutions.
7. **Activity variables and their inter relationship :** The activity variables refer to any activities which are in competition with other variables for limited resources, services, projects, products etc.

8. **Course of Action** : Different courses of action or alternatives should be available to a decision maker, who is required to make the decision which is the best effective or the optimal.
9. **Mutually exclusive criterion** : All decision parameters and the variables are assumed to be mutually exclusive. In other words, the occurrence of anyone variable rules out the simultaneous occurrence of other such variables.

IV. ADVANTAGES OF USING LINEAR PROGRAMMING

1. Linear programming technique helps decision-makers to use their productive resources effectively.
2. Linear programming technique improves the quality of decisions. The decision-making approach of the user of this technique becomes more objective and less subjective.
3. Linear programming technique helps to arrive at optimal solution of a decision problem by taking into account constraints on the use of resources. **For example**, saying that so many units of any product may be produced does not mean that all units can be sold.
4. Linear programming approach for solving decision problem highlight bottlenecks in the production processes. **For example** , when a bottleneck occurs, machine cannot produce sufficient number of units of a product to meet demand. Also, machines may remain idle.
5. It also helps in the re – evaluation of a basic plan for changing conditions.

V. LIMITATIONS OF LINEAR PROGRAMMING

1. Linear programming assumes linear relationships among decision variables. However, in real-life problems, decision variables, neither in the objective function nor in the constraints are linearly related.
2. While solving an LP model there is no guarantee that decision variables will get integer value. **For example**, how many men/machines would be required to perform a particular job, a non-integer valued solution will be meaningless. Rounding off the solution to the nearest integer will not yield an optimal solution.
3. The linear programming model does not take into consideration the effect of time and uncertainty.
4. For large – scale problems can be solved with LP techniques even when the assistance of a computer availability.
5. Parameters in the model are assumed to be constant but in real-life situations, they are frequently neither known nor constant.
6. Linear programming deals with only single objective, whereas in real-life situations a decision problem may have conflicting and multiple objectives.

VI. APPLICATION AREAS OF LINEAR PROGRAMMING

1. **Applications in Agriculture** : These applications fall into categories of farm economics and farm management. The former deals with interregional competition, optimum allocation of crop production, efficient production patterns under regional land resources and national demand constraints, while the latter is concerned with the problems of the individual farm such as allocation of limited resources such as acreage, labour, water supply, working capital, etc., so as to maximize the net revenue.

2. Applications in Military : Military applications include

- Selection of an air weapon system against the enemy,
- Ensuring minimum use of aviation gasoline
- Updating supply-chain to maximize the total tonnage of bombs dropped on a set of targets and takes care of the problem of community defence against disaster at the lowest possible cost.

3. Production Management

- **Product Mix** : To determine the quantity of several different products to be produced, knowing their per unit profit (cost) contribution and amount of limited production resources used. The objective is to maximize the total profit subject to all constraints.
- **Production Planning** : This deals with the determination of minimum cost production plan over the planning period, of an item with a fluctuating demand, while considering the initial number of units in inventory, production capacity, constraints on production, manpower and all relevant cost factors. The objective is to minimize total operation costs.
- **Assembly-line Balancing** : This problem is likely to arise when an item can be made by assembling different components. The process of assembling requires some specified sequence(s). The objective is to minimize the total elapse time.
- **Blending Problems** : These problems arise when a product can be made from a variety of available raw materials, each of which has a particular composition and price. The objective here is to determine the minimum cost blend, subject to availability of the raw materials, and to minimum and maximum constraints on certain product constituents.
- **Trim Loss** : When an item is made to a standard size (e.g. glass, paper sheet), the problem of determining which combination of requirements should be produced from standard materials in order to minimize the trim loss, arises.

4. Financial Management

- **Portfolio Selection** : This deals with the selection of specific investment activity among several other activities. The objective here is to find the allocation which maximizes the total expected return or minimizes risk under certain limitations.
- **Profit Planning** : This deals with the maximization of the profit margin from investment in plant facilities and equipment, cash in hand and inventory.

5. Marketing Management

- **Media Selection** : The linear programming technique helps in determining the advertising media mix so as to maximize the effective exposure, subject to limitation of budget, specified exposure rates to different market segments, specified minimum and maximum number of advertisements in various media.
- **Travelling Salesman Problem** : The salesman's problem is to find the shortest route from a given city to each of the specified cities and then returning to the original point of departure, provided no city would be visited twice during the tour. Such type of problems can be solved with the help of the modified assignment technique.
- **Physical Distribution** Linear programming determines the most economic and efficient manner of locating manufacturing plants and distribution centres for physical distribution.

6. Personnel Management

- **Staffing Problem** : Linear programming is used to allocate optimum manpower to a particular job so as to minimize the total overtime cost or total manpower.
- **Determination of Equitable Salaries** Linear programming technique has been used in determining equitable salaries and sales incentives.
- **Job Evaluation and Selection** : Selection of suitable person for a specified job and evaluation of job in organizations has been done with the help of the linear programming technique.

7. Other applications of linear programming lie in the area of administration, education, fleet utilization, awarding contracts, hospital administration, capital budgeting, etc.

VII. GENERAL MATHEMATICAL MODEL OF LINEAR PROGRAMMING PROBLEM

The general linear programming problem (or model) with n decision variables and m constraints can be stated in the following form:

- Optimize (Max. or Min.) $Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$
- Subject to the linear constraints,
 $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n (\leq, =, \geq) b_1$
 $a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n (\leq, =, \geq) b_2$
 $a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n (\leq, =, \geq) b_m$
 and $x_1, x_2, \dots, x_n \geq 0$

The above formulation can also be expressed in a compact form as follows.

$$\text{Optimize (Max. or Min.) } Z = \sum_{j=1}^n c_j x_j \quad (\text{Objective function}) \quad (1)$$

subject to the linear constraints

$$\sum_{j=1}^n a_{ij} x_j (\leq, =, \geq) b_i; \quad i=1, 2, 3, \dots, m \quad (\text{Constraints}) \quad (2)$$

$$\text{And } x_j \geq 0; \quad j = 1, 2, \dots, n \quad (\text{Non-negativity conditions}) \quad (3)$$

where, the c_j s are coefficients representing the per unit profit (or cost) of decision variable x_j to the value of objective function. The a_{ij} 's are referred as *technological coefficients (or input-output coefficients)*.

These represent the amount of resource, say i consumed per unit of variable (activity) x_j . These coefficients can be positive, negative or zero. The b_i represents the *total availability of the i th resource*. The term resource is used in a very general sense to include any numerical value associated with the right-hand side of a constraint. It is assumed that $b_i \geq 0$ for all i . However, if any $b_i < 0$, then both sides of constraint i is multiplied by -1 to make $b_i > 0$ and reverse the inequality of the constraint.

In the general LP problem, the expression $(\leq, =, \geq)$ means that in any specific problem each constraint may take only one of the three possible forms:

- Less than or equal to ($\leq$)
- Equal to ($=$)
- Greater than or equal to ($\geq$)

VIII. GUIDELINES ON LINEAR PROGRAMMING MODEL FORMULATION**Step 1: Identify the decision variables**

- Express each constraint in words. For this you should first see whether the constraint is of the form $\geq$ (at least as large as), of the form $\leq$ (no larger than) or of the form $=$ (exactly equal to).
- Express verbally the objective function.
- Verbally identify the decision variables with the help of Step (a) and (b). For this you need to ask yourself the question – *What decisions must be made in order to optimize the objective function?* Having followed Step 1(a) to (c) decides the symbolic notation for the decision variables and specify their units of measurement. Such specification of units of measurement would help in interpreting the final solution of the LP problem.

Step 2: Identify the problem data

- To formulate an LP model, identify the problem data in terms of constants, and parameters associated with decision variables. It may be noted that the decision-maker can control values of the variables but cannot control values in the data set.

Step 3: Formulate the constraints

- Convert the verbal expression of the constraints in terms of resource requirement and availability of each resource. Then express each of them as linear equality or inequality, in terms of the decision variables defined in Step 1. Values of these decision variables in the optimal LP problem solution must satisfy these constraints in order to constitute an acceptable (feasible) solution. Wrong formulation can either lead to a solution that is not feasible or to the exclusion of a solution that is actually feasible and possibly optimal.

Step 4: Formulate the objective function

- Identify whether the objective function is to be maximized or minimized. Then express it in the form of linear mathematical expression in terms of decision variables along with profit (cost) contributions associated with them.
- After gaining enough experience in model building, readers may skip verbal description. The following are certain examples of LP model formulation that may be used to strengthen the ability to translate a real life problem into a mathematical model.